LLM Transcript and Citation

Tool Used: ChatGPT (OpenAI, GPT-5.2)

Purpose: Assisting with structuring the manually created HTML and JSON files and resolving issues related to importing and comparing the datasets in R.

Course: DATA 607 – Assignment 7 (Working with HTML and JSON)

Citation:

ChatGPT (GPT-5.2) [Large language model]. https://chat.openai.com.

I used ChatGPT to assist with planning the structure of the HTML and JSON datasets, debugging the HTML table import process, and resolving a mismatch encountered during the data frame comparison step. The interactions below highlight the most relevant conversations that contributed to completing the assignment.

2. Conversation with ChatGPT (Creating the HTML and JSON files)

Prompt:

Should the `.html` and `.json` files be created in RStudio, and do they need to be written manually?

Assistant (ChatGPT):

Yes. The files can be created in RStudio as plain text files, but they should be typed manually rather than generated with R code. The HTML file should contain a table with the book attributes, and the JSON file should contain the same records as structured objects.

Suggested code / structure:

HTML file:

```
<table border="1">
  <tr>
    <th>title</th>
    <th>authors</th>
    <th>publication_year</th>
    <th>publisher</th>
    <th>genre</th>
  </tr>
</table>
```

JSON file:

```
[
  {
    "title": "An Introduction to Statistical Learning",
    "authors": "Gareth James, Daniela Witten, Trevor Hastie, Robert
Tibshirani"
  }
]
```

3. Conversation with ChatGPT (Resolving the Dataset Comparison Mismatch)

Prompt:

Why did `identical()` return `FALSE` even though the datasets looked the same?

Assistant (ChatGPT):

The mismatch was caused by a character encoding issue in the imported HTML author name, where accented characters did not match the JSON version. A clean fix was to standardize the author name text and then align both data frames before comparison.

Suggested code:

```
books_html <- books_html %>%
  mutate(across(everything(), as.character)) %>%
  as.data.frame(stringsAsFactors = FALSE)

books_json <- books_json %>%
  mutate(across(everything(), as.character)) %>%
  as.data.frame(stringsAsFactors = FALSE)

books_html <- books_html[, names(books_json), drop = FALSE]

row.names(books_html) <- NULL
row.names(books_json) <- NULL
```